

FINANCIAL MODEL & UNIT ECONOMICS

How much Vertil earns per subscriber, the cost to serve each plan, and the profit margin at typical and maximum usage — with a 12-month revenue model.

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1 · Assumptions & cost basis

Vertil runs on Anthropic's Claude API. Everyday generations use the cost-efficient Sonnet 4.6; an optional “Premium polish” uses Opus 4.8. The Nigerian voice lives in Vertil's prompts, so Sonnet still produces strong, on-brand output.

Model	Input \$/1M	Output \$/1M	Cost / generation	Cost / calendar
Opus 4.8 (premium)	\$5.00	\$25.00	₦66	₦210
Sonnet 4.6 (standard)	\$3.00	\$15.00	₦23	₦90
Haiku 4.5 (free tier)	\$1.00	\$5.00	₦8	₦30

A “generation” = one request returning 3 on-brand options (~1,200 input + ~700–1,400 output tokens). A calendar is a larger call (~3,500–5,000 output). Paystack fee = 1.5% + ₦100, capped at ₦2,000.

2 · Plan structure

Plan	Price / mo	Generations	Calendar	Default model	Notes
Free	₦0	10	—	Haiku 4.5	acquisition
Starter	₦7,500	150	—	Sonnet 4.6	1 brand voice
Growth □	₦19,000	600	Yes	Sonnet 4.6	3 brands · hero plan
Pro	₦45,000	Unlimited*	Yes	Sonnet + Opus	10 brands · *fair-use
Agency	₦150,000	Unlimited*	Yes	Sonnet + Opus	white-label · proposed

3 · Revenue & margin per subscriber

For each paid plan: what Vertil collects, the Paystack fee, the net, and the AI cost to serve — at maximum usage (every generation in the plan used) and at typical usage (~35% of the cap, the realistic average).

Plan	Price	Paystack fee	Net revenue	AI cost (max)	Profit (max)	Margin (max)
Starter	₦7,500	₦212	₦7,288	₦3,384	₦3,904	52%
Growth	₦19,000	₦385	₦18,615	₦14,075	₦4,540	24%
Pro	₦45,000	₦775	₦44,225	₦46,915	₦-2,690	-6%
Agency	₦150,000	₦2,000	₦148,000	₦118,186	₦29,814	20%

Plan	AI cost (typical)	Profit (typical)	Margin (typical)	Break-even	Verdict
Starter	₦1,184	₦6,103	81%	323 gens	Cap below break-even — always profitable

Plan	AI cost (typical)	Profit (typical)	Margin (typical)	Break-even	Verdict
Growth	₦4,926	₦13,689	72%	825 gens	Cap below break-even — always profitable
Pro	₦16,420	₦27,805	62%	1,960 gens	Cap ABOVE break-even — needs fair-use guard
Agency	₦41,365	₦106,635	71%	6,560 gens	Cap below break-even — always profitable

Reading this

- Starter & Growth are bullet-proof. Their generation caps (150 and 600) sit below the break-even point, so even a user who maxes out every month is profitable — Growth still nets ~24% at full tilt and ~72% at typical use.
- Pro is the one to watch. “Unlimited” breaks even around ~1,960 generations/month; a true power-user past that erodes the margin. Two guards keep it safe: a fair-use ceiling (~1,500–2,000/mo) and keeping Sonnet — not Opus — as the default. If Pro ran every generation on Opus, cost per generation triples (₦66 vs ₦23) and heavy use goes negative.
- Agency is the highest-margin line at typical use (~71%), with near-zero marginal cost beyond AI — and the lowest churn, because agencies build their own business on top of it.

4 • The cost of Free (customer acquisition)

A Free user costs about ₦75/month to serve (10 generations on Haiku). At 8,000 free users that's ~₦601,600/month — the true cost of acquisition, recovered the moment ~6% convert to paid. Running Free on Haiku (not Sonnet) keeps this ~3× cheaper.

5 • Contribution per paid subscriber (typical use)

What each subscriber contributes to gross profit in an average month — the number that compounds.

Plan	Net revenue	AI cost	Gross profit / mo	Gross profit / yr
Starter	₦7,288	₦1,184	₦6,103	₦73,237
Growth	₦18,615	₦4,926	₦13,689	₦164,267
Pro	₦44,225	₦16,420	₦27,805	₦333,656
Agency	₦148,000	₦41,365	₦106,635	₦1,279,620

6 • 12-month revenue model

Paid mix Starter 45% / Growth 35% / Pro 15% / Agency 5% → blended ARPPU ≈ ₦24,300/mo. Gross margin held at ~70% (Sonnet default, typical utilization). Monthly churn ~6% (base).

Scenario (month 12)	Free users	Conv.	Paid users	MRR	ARR	Gross profit / mo
Conservative	3,000	4%	120	₦2,913,000	₦34,956,000	₦2,039,100
Base	8,000	6%	480	₦11,652,000	₦139,824,000	₦8,156,400
Optimistic	20,000	8%	1,600	₦38,840,000	₦466,080,000	₦27,188,000

Blended ARPPU = ₦24,275 / paying user / month. Base case ≈ ₦139,824,000 ARR (~\$87,390) by month 12.

7 · Key risks & the levers that protect margin

- Model tiering is non-negotiable. Sonnet default keeps gross margin ~70–85%; all-Opus would turn heavy users loss-making. Reserve Opus for a paid “Premium polish”.
- Fair-use ceiling on “unlimited” Pro/Agency (~1,800–2,000 generations) caps the tail risk while staying invisible to 99% of users.
- Annual plans (2 months free) convert monthly cancel-decisions into yearly ones and front-load cash — the single biggest churn lever.
- FX exposure. Costs are USD (Claude API); revenue is Naira. A weaker Naira raises cost per generation — model tiering is the hedge.
- Prompt caching can cut input-token cost up to ~90% on the repeated brand/system prefix, improving every margin above.

TL;DR

- Per subscriber (typical use): Starter ~81% margin, Growth ~72%, Pro ~62%, Agency ~71%.
- At max usage: Starter ~52% and Growth ~24% stay profitable; Pro needs a fair-use ceiling.
- Margins only work with model tiering (Sonnet default, Opus as a premium add-on).
- Base case: ~~~N~~11,652,000 MRR by month 12 at ~70% gross margin.